The **University Of Azad Jammu and Kashmir, Muzaffarabad**



Department of Software Engineering

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Course Instructor	**Engr. Seemab Khurshid**
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Session	**2025–29**
Lab No	**01**
Lab Title	**Introduction to Operating Systems**
Due Date	**20-Oct-2026**

1. Define an operating system and explain its role in managing computer hardware and software.

An operating system is system software that acts as a link between the user, application programs, and computer hardware. It helps the computer work properly by managing its different resources.
The operating system manages the processor, memory, storage devices, and other hardware components.

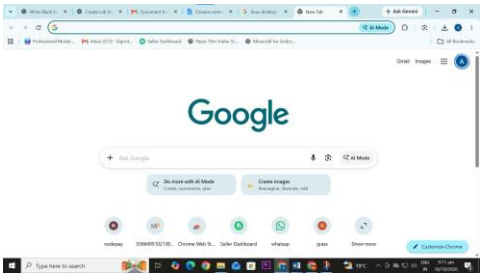
Examples: Microsoft Windows, Linux, macOS, and Android.



2. Differentiate between system software and application software with suitable examples.

System software controls the computer and provides the basic environment in which other programs can run. Application software is designed to help us perform specific tasks.

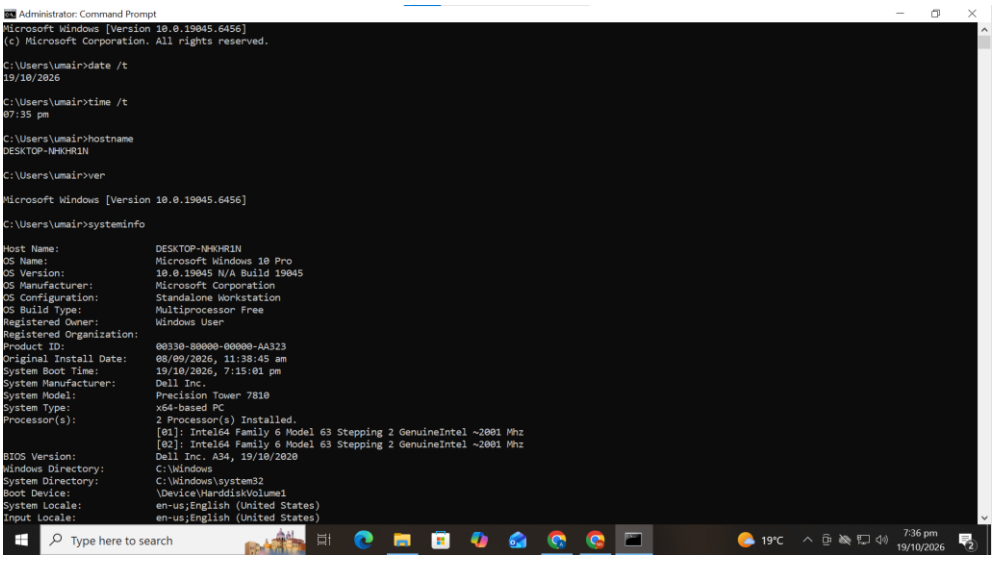
System Software	Application Software
• It manages computer hardware and basic system operations.	• It helps users perform particular tasks.
• It provides a platform for applications to run.	• It runs on top of the operating system.
• Examples include Windows, Linux, and device drivers	• Examples include Microsoft Word, Google Chrome, and Adobe Photoshop.
• It mainly supports the operation of the computer.	• It mainly helps users complete their work.

System Software	**Application Software**
	
Windows	Chrome

3. Write DOS commands to display the current date, time, computer name, and operating system version.

I can use Windows Command Prompt (CMD) to display basic information about my computer. The following commands help me check the current date and time, identify my computer, and view the Windows version.

Information	Command	Purpose
Current date	date /t	Displays the current date.
Current time	time /t	Displays the current time.
Computer name	hostname	Displays the computer's name.
Windows version	ver	Displays the Windows version information.
Detailed system information	systeminfo	Displays information about Windows and the computer's hardware.

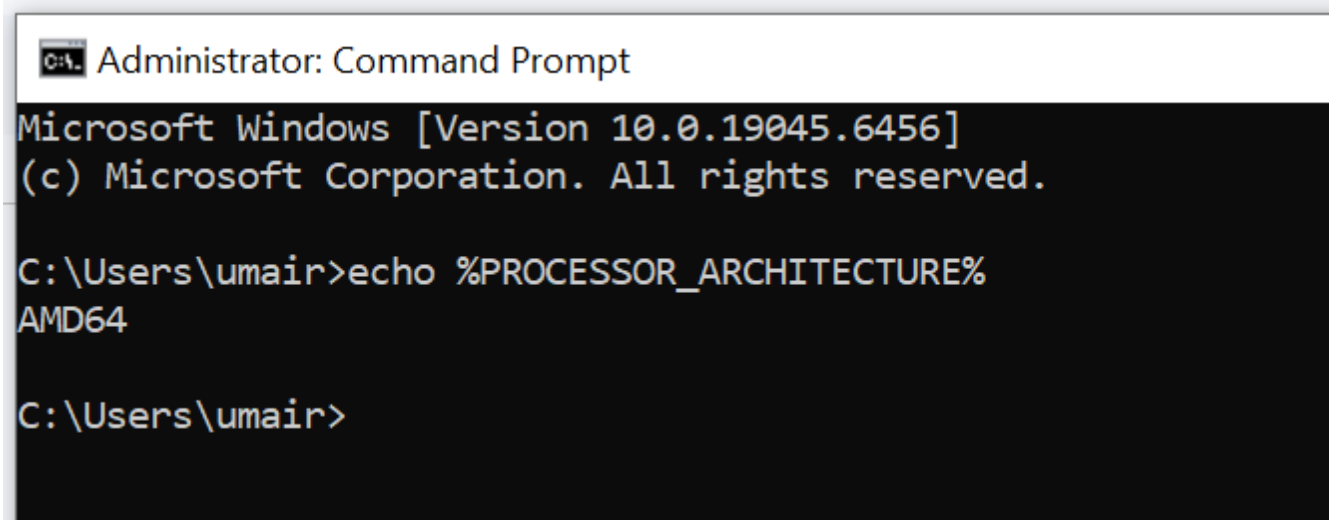


4. Identify the processor architecture (32-bit or 64-bit) of your system using Windows Command Prompt.

Processor architecture tells us whether the system is using a 32-bit or 64-bit architecture. A 64-bit system can handle 64-bit operating systems and applications and can support more memory than a typical 32-bit system.

I can check the processor architecture using Windows Command Prompt with the following command:

I opened Command Prompt, entered the architecture command, (echo %PROCESSOR_ARCHITECTURE%) and checked the result to identify the processor architecture.

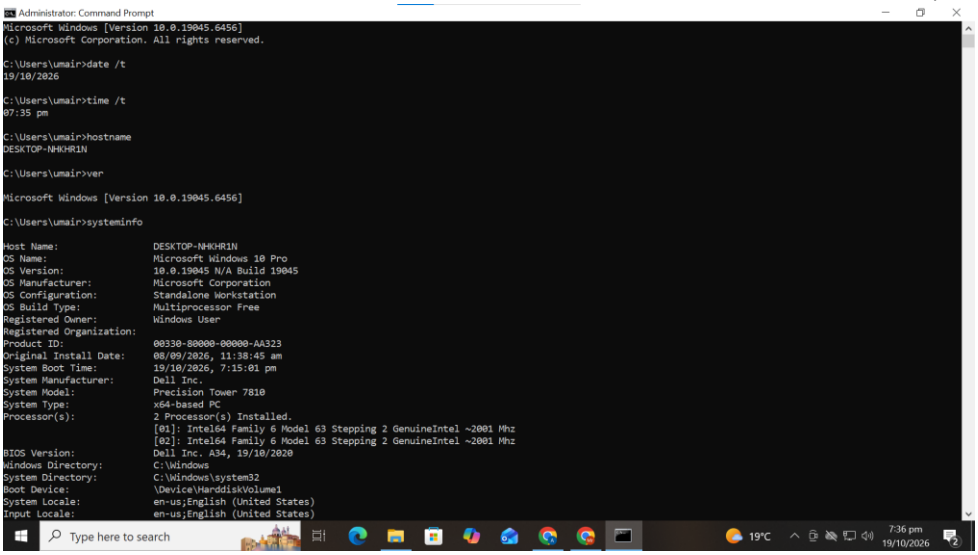


5. Execute appropriate DOS and Linux commands to display basic system information.

Windows Command Prompt and the Linux terminal both provide commands for checking system information. I can use these commands to learn about the operating system, computer name, processor architecture, and hardware.

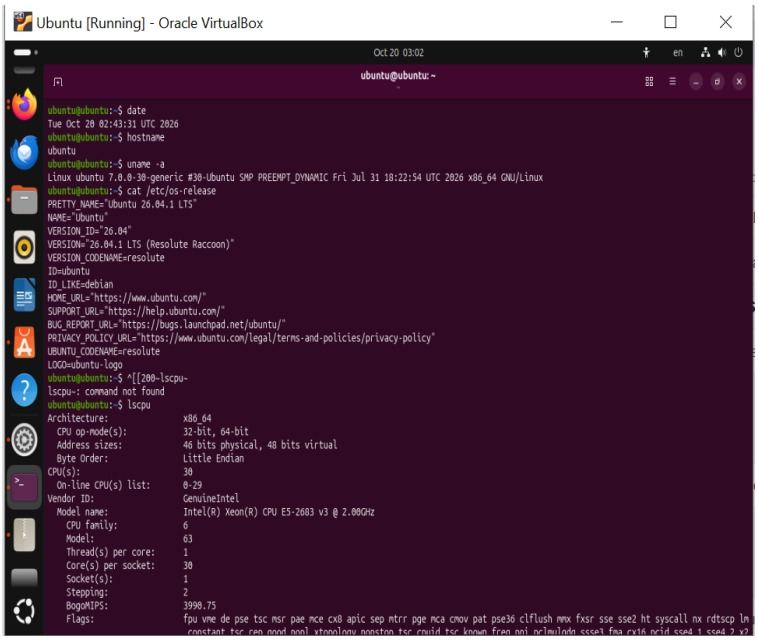
A. Windows Command Prompt

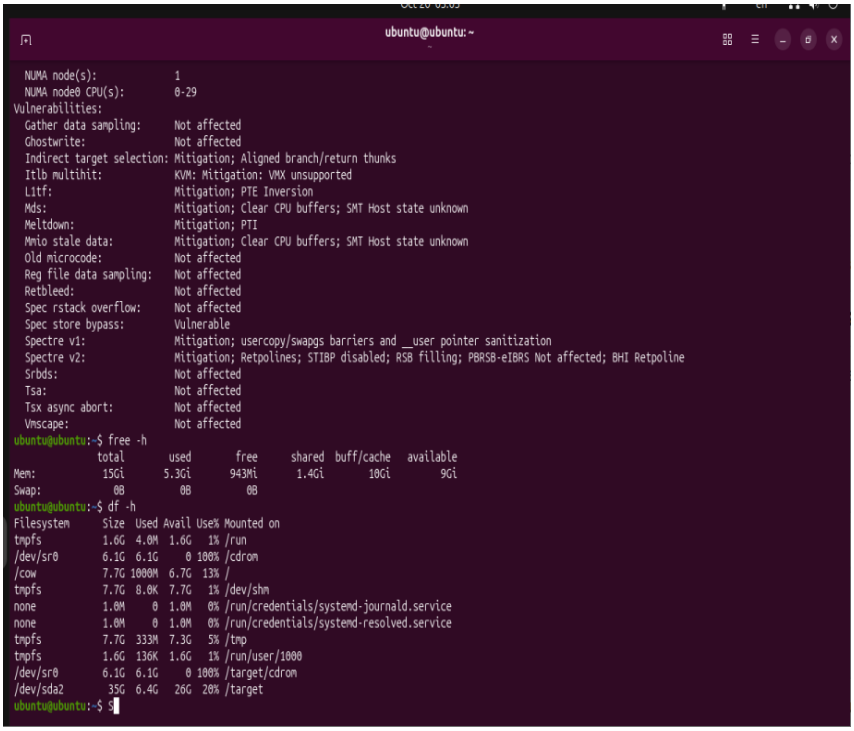
Command	Information displayed
`date /t`	Current date
`time /t`	Current time
`hostname`	Computer name
`ver`	Windows version
`systeminfo`	Detailed system information



B. Linux Terminal

Command	Information displayed
`date`	Current date and time
`hostname`	Computer name
`uname -a`	Kernel and general system information
`cat /etc/os-release`	Linux distribution and version information
`lscpu`	Processor architecture, CPU model, and CPU details
`free -h`	Memory usage
`df -h`	Disk space usage





6. Compare the commands used in Windows Command Prompt and the Linux terminal for retrieving system information.

Windows Command Prompt and the Linux terminal both allow users to retrieve system information through text commands. However, their commands and output formats are different.

Purpose	**Windows CMD**	**Linux Terminal**
Display date and time	`date /t` and `time /t`	`date`
Display computer name	`hostname`	`hostname`
Display OS or kernel information	`ver` or `systeminfo`	`uname -a`
Display OS distribution details	`systeminfo`	`cat /etc/os-release`
Display CPU information	`systeminfo` or `wmic cpu get name` on systems that support WMIC	`lscpu`
Display memory information	`systeminfo`	`free -h`
Display disk space	`wmic logicaldisk get size,freespace,caption` on systems that support WMIC	`df -h`

In my understanding, both command-line environments perform similar tasks, but they use different commands. Windows CMD provides commands such as `ver` and `systeminfo`, while Linux uses commands such as `uname`, `lscpu`, and `free`. Linux also offers many small commands that each display a particular type of information. The main difference is the command syntax and the way information is presented.

Identify CPU Cores and Processor Clock Speed

1. Define a CPU core and differentiate between physical cores and logical processors.

A CPU core is a processing unit inside a processor that performs calculations and executes instructions. A processor can contain multiple cores, which help the computer handle several tasks efficiently.

Physical cores are the actual processing units built into a CPU. Logical processors are the processing units that the operating system sees and uses to schedule tasks. Some physical cores can handle two logical processing threads through technologies such as Intel Hyper-Threading.

For example, a processor with 2 physical cores and 4 logical processors has two actual cores that can handle four logical execution threads.

2. Write a DOS command to display processor-related information.

I used Windows Command Prompt to check the processor information available on my computer. The following commands can be used:

Command	Purpose
`systeminfo`	Displays system information, including processor details.
`wmic cpu get Name,NumberOfCores,NumberOfLogicalProcessors,MaxClockSpeed`	Displays the processor name, core count, logical processor count, and maximum clock speed on systems that support WMIC.
`echo %PROCESSOR_ARCHITECTURE%`	Displays the processor architecture reported by the command environment.

I opened Command Prompt, entered the commands, and observed the information displayed about the processor.

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\umair>systeminfo

Host Name:                 DESKTOP-NHKHR1N
OS Name:                   Microsoft Windows 10 Pro
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:          Windows User
Registered Organization:
Product ID:                00330-80000-00000-AA323
Original Install Date:     08/09/2026, 11:38:45 am
System Boot Time:          20/10/2026, 8:21:11 am
System Manufacturer:       Dell Inc.
System Model:              Precision Tower 7810
System Type:               x64-based PC
Processor(s):              2 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 63 Stepping 2 GenuineIntel ~2001 Mhz
                           [02]: Intel64 Family 6 Model 63 Stepping 2 GenuineIntel ~2001 Mhz
BIOS Version:              Dell Inc. A34, 19/10/2020
Windows Directory:         C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:             en-us;English (United States)
Input Locale:              en-us;English (United States)
Time Zone:                 (UTC-08:00) Pacific Time (US & Canada)
Total Physical Memory:     32,688 MB
Available Physical Memory: 20,620 MB
Virtual Memory: Max Size:  37,552 MB
Virtual Memory: Available: 22,965 MB
Virtual Memory: In Use:    14,587 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\DESKTOP-NHKHR1N
Hotfix(s):                 14 Hotfix(s) Installed.
                           [01]: KB5066130
                           [02]: KB5066135
                           [03]: KB5011048
```

```
C:\Users\umair>wmic cpu get Name,NumberOfCores,NumberOfLogicalProcessors,MaxClockSpeed
MaxClockSpeed  Name                                      NumberOfCores  NumberOfLogicalProcessors
2001           Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.00GHz  14             28
2001           Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.00GHz  14             28

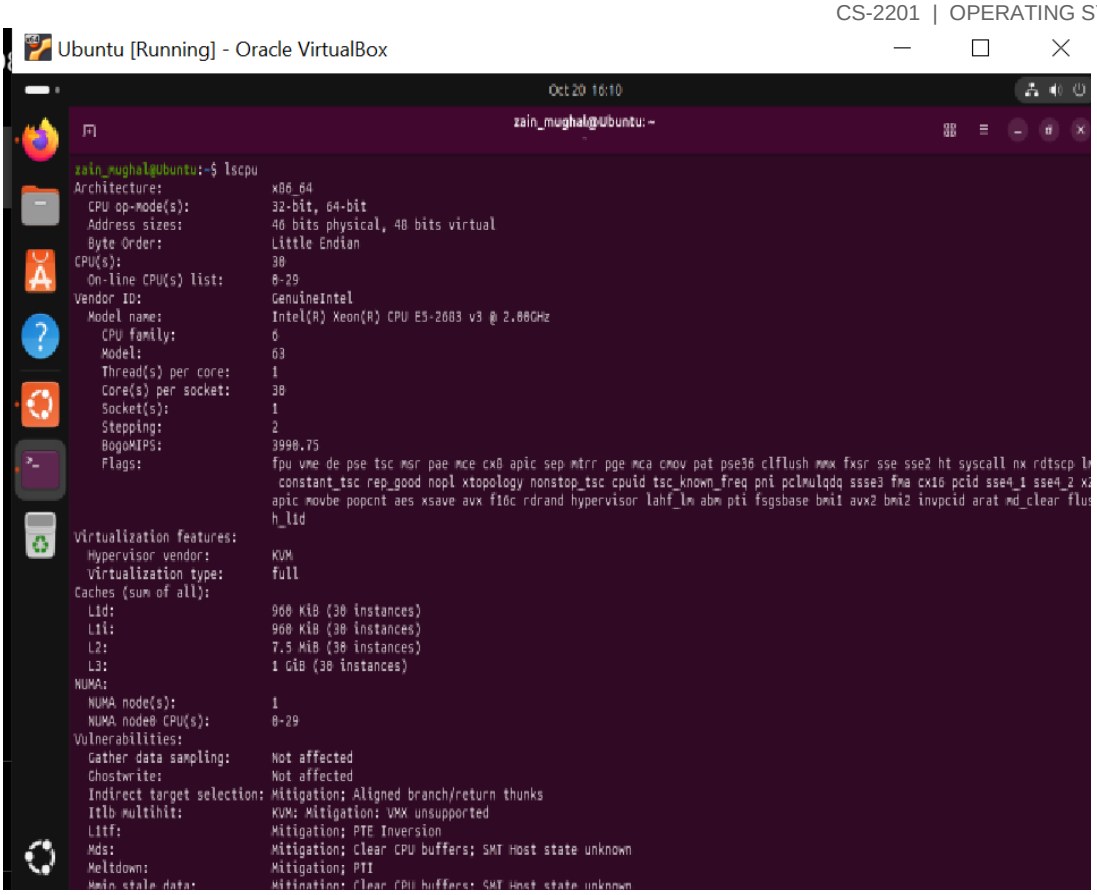

C:\Users\umair>echo %PROCESSOR_ARCHITECTURE%
AMD64
```

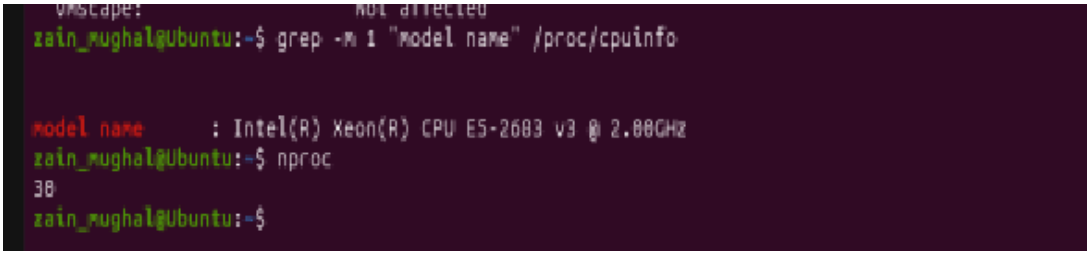
3. Use Linux shell commands to display the processor model, number of CPU cores, and CPU architecture.

Linux provides several commands for checking processor information. I can use the following commands in a Linux terminal:

Command	Purpose
`lscpu`	Displays CPU architecture, model, core information, and logical processor count.
`grep -m 1 "model name" /proc/cpuinfo`	Displays the processor model name.
`nproc`	Displays the number of processing units available to the current environment.

I opened the Linux terminal and executed these commands to identify the processor model, architecture, and available processing units.





The `lscpu` command is especially useful because it displays several important CPU details in one output.

4. Execute appropriate commands in Linux to display CPU information and processor details.

I used Linux commands to examine the processor configuration and its reported clock speed.

`Lscpu`

This command displays the processor architecture, model name, CPU count, cores, and other details.

```
zain_mughal@Ubuntu:~$ lscpu
Architecture:             x86_64
  CPU op-mode(s):         32-bit, 64-bit
  Address sizes:          46 bits physical, 48 bits virtual
  Byte Order:             Little Endian
CPU(s):                   30
  On-line CPU(s) list:    0-29
Vendor ID:                GenuineIntel
  Model name:             Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.80GHz
    CPU family:           6
    Model:                63
    Thread(s) per core:   1
    Core(s) per socket:   30
    Socket(s):            1
    Stepping:             2
    BogoMIPS:             3998.75
    Flags:                fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm
                           constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2
                          apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm pti fsgsbase bmi1 avx2 bmi2 invpcid arat md_clear flus
                          h_l1d
Virtualization features:
  Hypervisor vendor:      KVM
  Virtualization type:    full
Caches (sum of all):
  L1d:                    960 KiB (30 instances)
  L1i:                    960 KiB (30 instances)
  L2:                     7.5 MiB (30 instances)
  L3:                     1 GiB (30 instances)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-29
Vulnerabilities:
  Gather data sampling:   Not affected
  Ghostwrite:             Not affected
  Indirect target selection: Mitigation; Aligned branch/return thunks
  Itlb multihit:          KVM: Mitigation: VMX unsupported
  Litf:                   Mitigation; PTE Inversion
  Mds:                    Mitigation; Clear CPU buffers; SMT Host state unknown
  Meltdown:               Mitigation; PTI
```

`Cat /proc/cpuinfo`

This command displays detailed information about the logical processors recognized by Linux.

```
cat: command not found
zain_mughal@Ubuntu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 63
model name      : Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.80GHz
stepping        : 2
microcode       : 0x43
cpu MHz         : 1995.378
cache size      : 35840 KB
physical id     : 0
siblings        : 30
core id         : 0
cpu cores       : 30
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 15
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_t
sc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdr
and hypervisor lahf_lm abm pti fsgsbase bmi1 avx2 bmi2 invpcid arat md_clear flush_l1d
bugs            : cpu_meltdown spectre_v1 spectre_v2 spec_store_bypass l1tf mds swapgs itlb_multihit mmio_stale_data bhi spectre_v2_user its
bogomips        : 3998.75
clflush size    : 64
cache_alignment : 64
address sizes   : 46 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id       : GenuineIntel
cpu family      : 6
model           : 63
model name      : Intel(R) Xeon(R) CPU E5-2683 v3 @ 2.80GHz
stepping        : 2
microcode       : 0x43
cpu MHz         : 1995.378
cache size      : 35840 KB
```

`lscpu | grep -i "mhz"`

This command displays clock-speed fields reported by `lscpu`, if available.



`Nproc`

This command displays the number of processing units available to the current environment.

I ran these commands and examined their outputs. I used `lscpu` to get a summary of the CPU and `/proc/cpuinfo` to view more detailed processor information.

5. Explain the difference between CPU clock speed and the number of CPU cores.

CPU clock speed refers to the frequency at which a processor operates. It is usually measured in gigahertz (GHz). For example, 2.40 GHz represents approximately 2.4 billion clock cycles per second.

The number of CPU cores refers to the number of physical processing units inside a processor. More cores can improve performance when a program can divide its work between them.

CPU clock speed	Number of CPU cores
Measured in GHz.	Measured as a count of cores.
Indicates the processor's operating frequency.	Indicates the number of physical processing units.
Can change according to workload, temperature, and power settings.	Normally remains fixed for a particular processor.
Example: 2.40 GHz.	Example: 2 physical cores.

In my understanding, clock speed and core count describe different aspects of processor performance. A processor with a higher clock speed may perform some tasks faster, while a processor with more cores may handle multiple tasks more efficiently. Actual performance also depends on processor design, software, memory, and workload.

Explore Task Manager and Process Management

1. Write a DOS command to display all currently running processes.

Windows Command Prompt provides the `tasklist` command to display the processes currently running on the computer. It shows details such as process names, process IDs (PIDs), and memory usage.

Command:

`Tasklist`

I opened Command Prompt, entered `tasklist`, and viewed the list of running processes on my computer.

```
C:\Users\umair>tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0      1,504 K
Registry                       424 Services                   0     83,448 K
smss.exe                      1020 Services                   0        936 K
csrss.exe                     1100 Services                   0      5,516 K
wininit.exe                   1188 Services                   0      6,192 K
csrss.exe                     1196 Console                    1      5,876 K
services.exe                  1260 Services                   0      9,312 K
lsass.exe                     1284 Services                   0     21,600 K
winlogon.exe                  1352 Console                    1     10,088 K
svchost.exe                   1476 Services                   0     27,124 K
fontdrvhost.exe               1500 Services                   0      3,064 K
fontdrvhost.exe               1508 Console                    1      5,304 K
svchost.exe                   1604 Services                   0     15,628 K
svchost.exe                   1648 Services                   0      9,988 K
svchost.exe                   1728 Services                   0      5,092 K
svchost.exe                   1760 Services                   0      8,996 K
svchost.exe                   1768 Services                   0     11,840 K
svchost.exe                   1844 Services                   0      6,036 K
svchost.exe                   1876 Services                   0     14,636 K
svchost.exe                   2004 Services                   0      6,484 K
dwm.exe                       2040 Console                    1     72,600 K
svchost.exe                   1068 Services                   0      7,420 K
WUDFHost.exe                  1416 Services                   0      7,432 K
NVDisplay.Container.exe       1304 Services                   0     27,464 K
svchost.exe                   2084 Services                   0     14,364 K
svchost.exe                   2092 Services                   0     11,200 K
svchost.exe                   2228 Services                   0     13,628 K
svchost.exe                   2236 Services                   0      5,756 K
svchost.exe                   2244 Services                   0     12,592 K
svchost.exe                   2252 Services                   0      9,996 K
svchost.exe                   2260 Services                   0      6,916 K
svchost.exe                   2432 Services                   0      7,200 K
Memory Compression            2448 Services                   0    369,932 K
svchost.exe                   2512 Services                   0      6,064 K
```

2. Use Windows Command Prompt to identify a running process by its name or PID.

I can use the `tasklist` command to search for a particular process by its name or process ID.

To find a process by name:

```
tasklist | findstr /i "chrome"
```

This command searches the process list for entries containing `chrome`, without distinguishing between uppercase and lowercase letters.

```
C:\Users\umair>tasklist | findstr /i "chrome"
chrome.exe                    7608 Console                    1    383,948 K
chrome.exe                    5960 Console                    1      8,696 K
chrome.exe                   12604 Console                    1    224,992 K
chrome.exe                   12560 Console                    1     79,428 K
chrome.exe                   12780 Console                    1     19,952 K
chrome.exe                   13408 Console                    1    108,124 K
chrome.exe                   11456 Console                    1     64,716 K
chrome.exe                    5968 Console                    1    581,224 K
chrome.exe                   11956 Console                    1     49,732 K
chrome.exe                   14796 Console                    1     23,760 K
chrome.exe                   14804 Console                    1     35,244 K
chrome.exe                   13712 Console                    1     46,700 K
chrome.exe                    2908 Console                    1     49,040 K
chrome.exe                   11184 Console                    1     92,632 K
chrome.exe                    8924 Console                    1    494,380 K
chrome.exe                   15608 Console                    1     45,752 K
chrome.exe                    8280 Console                    1     52,136 K
chrome.exe                    2608 Console                    1     73,888 K
chrome.exe                   13968 Console                    1    118,628 K
chrome.exe                   11492 Console                    1     44,108 K
chrome.exe                    9904 Console                    1    139,620 K
chrome.exe                   11916 Console                    1     52,196 K
chrome.exe                    3256 Console                    1     76,400 K
chrome.exe                    6152 Console                    1     70,020 K
chrome.exe                    8644 Console                    1     50,348 K
chrome.exe                    6068 Console                    1    122,396 K
chrome.exe                   10940 Console                    1     44,640 K
chrome.exe                    1832 Console                    1    175,464 K
chrome.exe                    7248 Console                    1     98,156 K
chrome.exe                   13936 Console                    1     55,872 K
chrome.exe                   11244 Console                    1    230,540 K
chrome.exe                   14580 Console                    1    220,184 K
chrome.exe                   16500 Console                    1     44,496 K
chrome.exe                    1408 Console                    1    306,372 K
chrome.exe                    8880 Console                    1    444,168 K
chrome.exe                   15320 Console                    1    416,444 K
chrome.exe                   17468 Console                    1     92,028 K
```

To find a process by PID:

`tasklist /FI "PID eq 1234"`

Here, `1234` is an example PID. I should replace it with the actual PID displayed on my computer.

tasklist /FI "PID eq 7608"

I searched for a running process by its name and then used its PID to identify the specific process.

```
C:\Users\umair>tasklist /FI "PID eq 7608"

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
chrome.exe                    7608 Console                    1    383,692 K
```

3. Write a DOS command to terminate a process using its PID.

Windows Command Prompt provides the `taskkill` command to terminate a process using its PID.

Command:

`taskkill /PID 1234 /F`

- `/PID 1234` specifies the process ID.
- `/F` forces the process to terminate.

I learned how to use `taskkill` to stop a process by its PID. I would first identify the correct process and verify its PID before executing the command.

taskkill /PID 7608 /F

```
C:\Users\umair>taskkill /PID 7608 /F
SUCCESS: The process with PID 7608 has been terminated.
```

4. Write Linux shell commands to display running processes and their resource usage.

Linux provides commands to view running processes and monitor the resources they use.

Command	Purpose
`ps aux`	Displays a detailed list of processes.
`top`	Shows running processes and updates resource usage in real time.
`free -h`	Displays memory usage in a readable format.

I used `ps aux` to view running processes and `top` to observe CPU and memory usage. These commands helped me understand how Linux monitors active processes and their resource consumption.

5. Execute a Linux command to identify a process using its name.

Linux provides the `pgrep` command to search for processes by name.

Command:

`pgrep -a firefox`

This command displays the PIDs and command lines of processes matching `firefox`, if any are running.

I can replace `firefox` with another process name, such as `chrome`, depending on the applications running on my system.

I used `pgrep -a` to search for a process by its name and identify its PID.

6. Write a Linux command to terminate a process using its PID.

Linux provides the `kill` command to send a signal to a process using its PID.

Command:

```
kill 1234
```

Here, `1234` is an example PID. The command normally sends the `TERM` signal, which asks the process to terminate gracefully.

If the process does not respond, a forceful termination can be requested with:

```
kill -9 1234
```

I should use the forceful option only when necessary, because it does not allow the process to shut down gracefully.

I learned how to identify a process, check its PID, and use the `kill` command to request its termination.

Explore Startup Applications and System Services

1. Explain the purpose of startup applications in an operating system.

Startup applications are programs that start automatically when the operating system boots or when a user signs in. They help make frequently used applications and background utilities available without opening them manually.

For example, security software, cloud-sync applications, and some communication tools may start automatically with Windows. However, having too many startup applications can increase boot time and use additional memory and CPU resources.

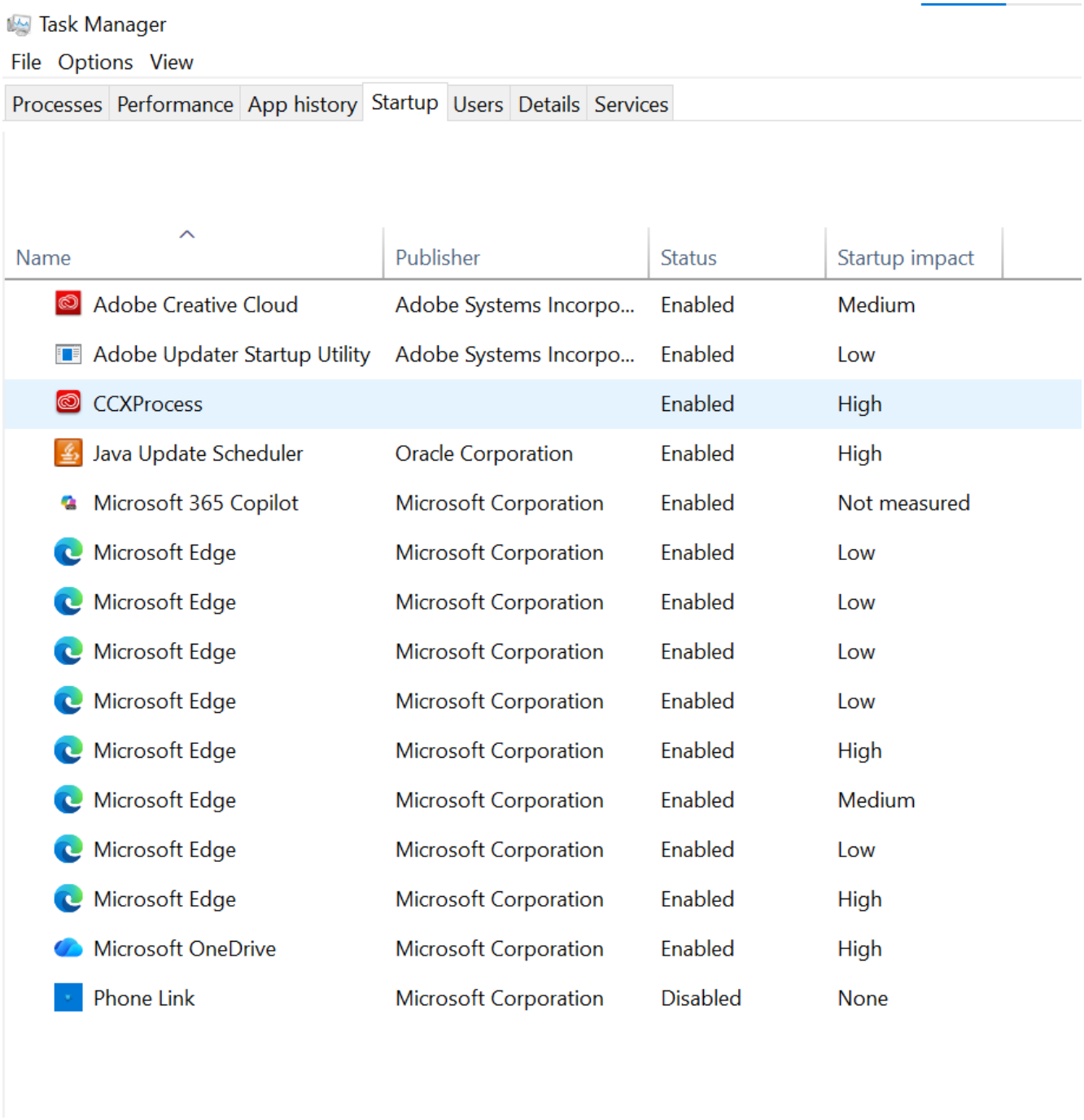
In my practical work, I explored the Startup section in Task Manager to understand how startup applications are listed and how their status and startup impact are displayed.

2. Identify the Windows tools used to examine startup applications and their impact on system startup.

Windows provides several tools for checking startup applications and managing system startup.

Windows tool	Purpose
Task Manager	Displays startup applications, their enabled or disabled status, and startup impact.
Settings	Allows users to enable or disable supported startup applications.
System Configuration (`msconfig`)	Provides access to selected system configuration and startup-related settings.

I opened Task Manager, selected the Startup tab, and reviewed the listed applications, their status, and their startup impact. This helped me understand which programs may affect the time required for Windows to start.



Name	Publisher	Status	Startup impact
Adobe Creative Cloud	Adobe Systems Incorpo...	Enabled	Medium
Adobe Updater Startup Utility	Adobe Systems Incorpo...	Enabled	Low
CCXProcess		Enabled	High
Java Update Scheduler	Oracle Corporation	Enabled	High
Microsoft 365 Copilot	Microsoft Corporation	Enabled	Not measured
Microsoft Edge	Microsoft Corporation	Enabled	Low
Microsoft Edge	Microsoft Corporation	Enabled	Low
Microsoft Edge	Microsoft Corporation	Enabled	Low
Microsoft Edge	Microsoft Corporation	Enabled	Low
Microsoft Edge	Microsoft Corporation	Enabled	High
Microsoft Edge	Microsoft Corporation	Enabled	Medium
Microsoft Edge	Microsoft Corporation	Enabled	Low
Microsoft Edge	Microsoft Corporation	Enabled	High
Microsoft OneDrive	Microsoft Corporation	Enabled	High
Phone Link	Microsoft Corporation	Disabled	None

3. Write a Linux command to list currently running services on a system using systemd.

Linux systems that use `systemd` can display currently running services with the following command:

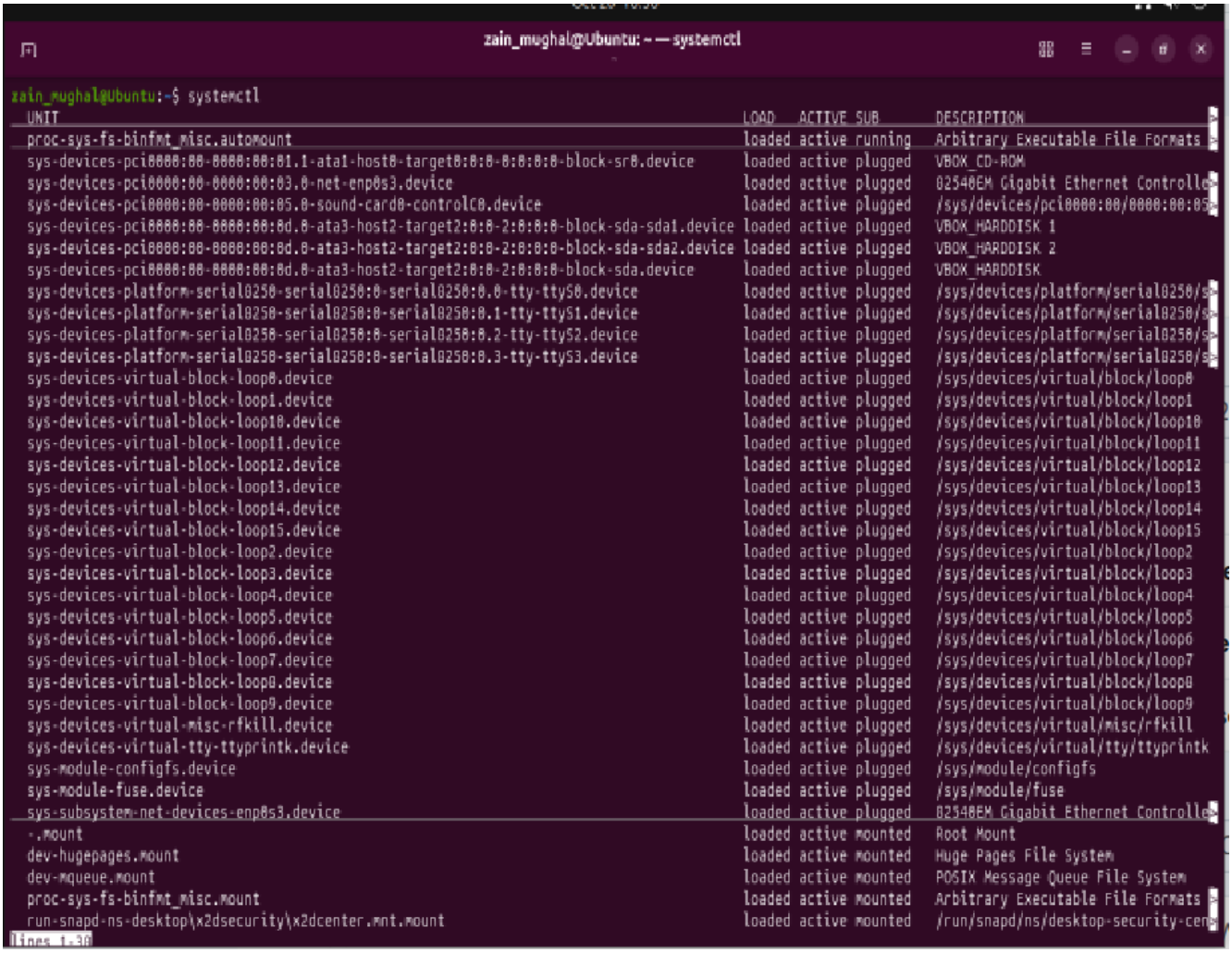
```
systemctl list-units --type=service --state=running
```

Explanation:

- `systemctl` is used to manage and inspect systemd services.
- `list-units` displays units currently loaded by systemd.

- `--type=service` limits the results to services.
- `--state=running` displays services that are currently running.

I opened the Linux terminal and executed this command to view the services currently running on the system.

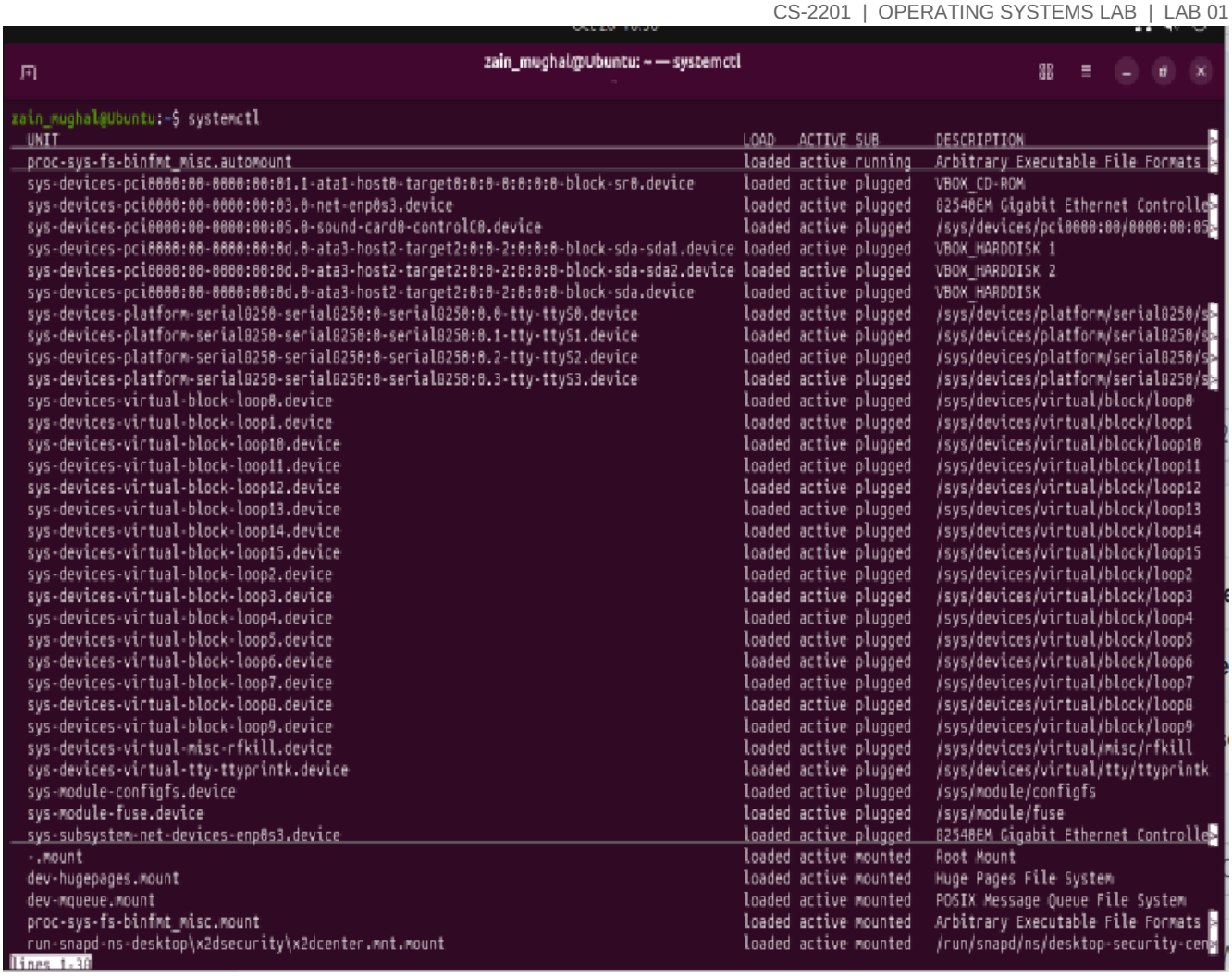


4. Write a Linux command to check the status of a specified service.

The `systemctl status` command displays information about a particular service, including its current state and recent log messages.

Command:

```
systemctl status ssh
```



This command checks the status of the service named `ssh` on systems where that service exists under that name.

On some Linux distributions, the SSH server service is named `sshd`, so I can use:

```
systemctl status sshd
```

I entered the appropriate command in the Linux terminal to check the service status. The output can show whether the service is active, inactive, failed, or unavailable.

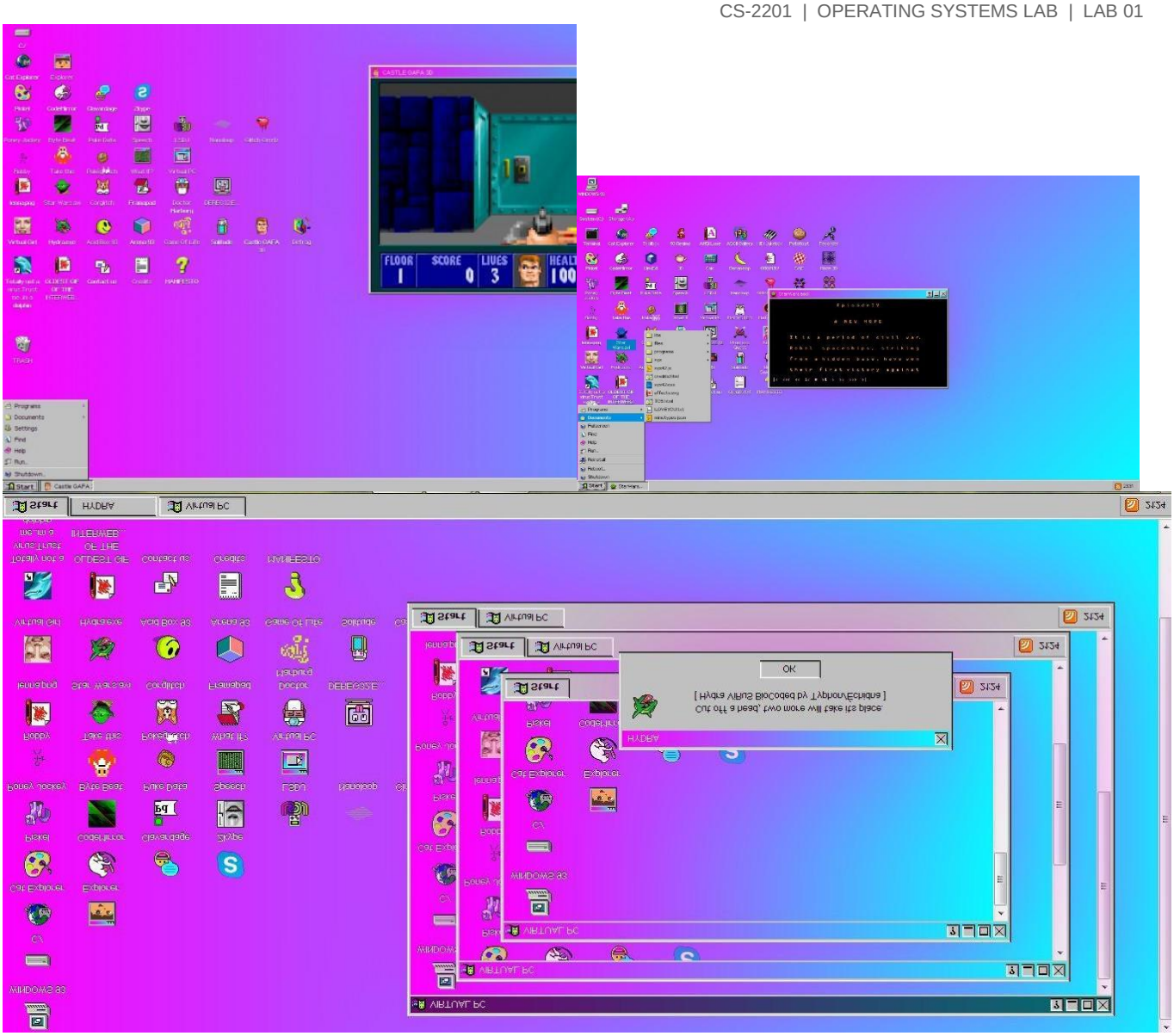
Explore Windows 93 and Windows 98

1. Explore a browser-based Windows 93 environment and identify its main desktop components

Objective: To explore the Windows 93 browser-based environment and identify its main desktop components.

Procedure:

1. I opened a web browser and visited the Windows 93 website: https://www.windows93.net/.
2. I explored the desktop interface and observed its different components.
3. I clicked on desktop icons and explored the available applications and menus.
4. I observed how the interface resembles an older Windows operating system.



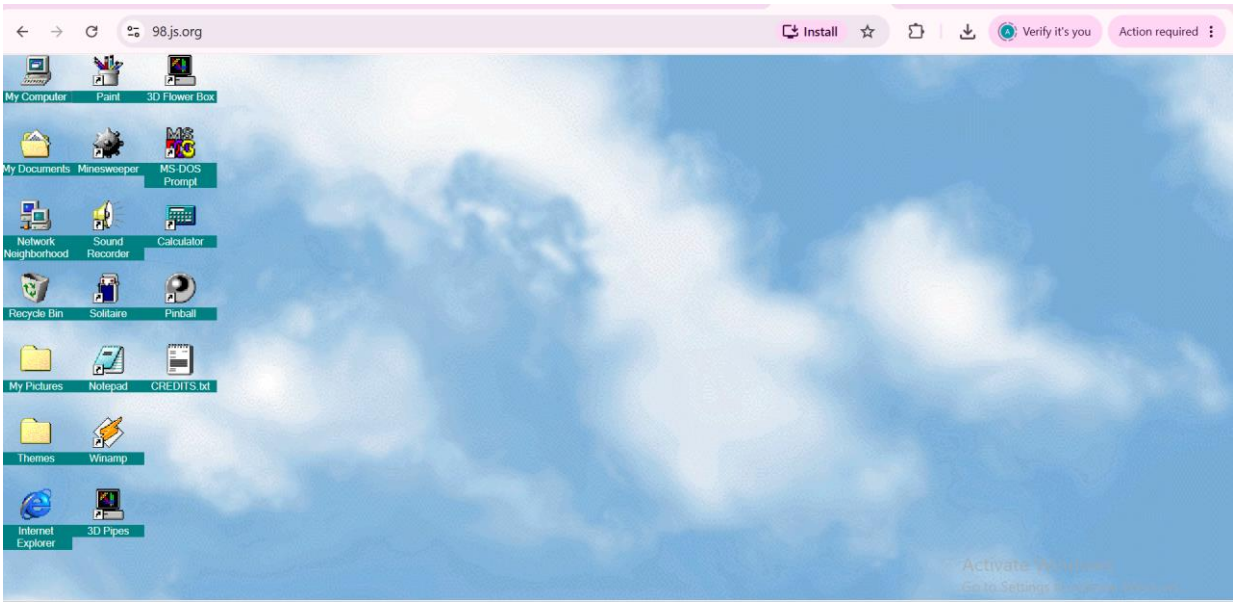
Main desktop components

Component	Description
Desktop	The main workspace where icons and applications appear.
Desktop icons	Shortcuts used to open applications, games, and other features.
Taskbar	A bar used to access open applications and desktop controls.
Start menu	A menu that provides access to applications and system options, where available.
Application windows	Windows that open when an application is launched.
Mouse pointer	Used to select icons, open menus, and interact with applications.

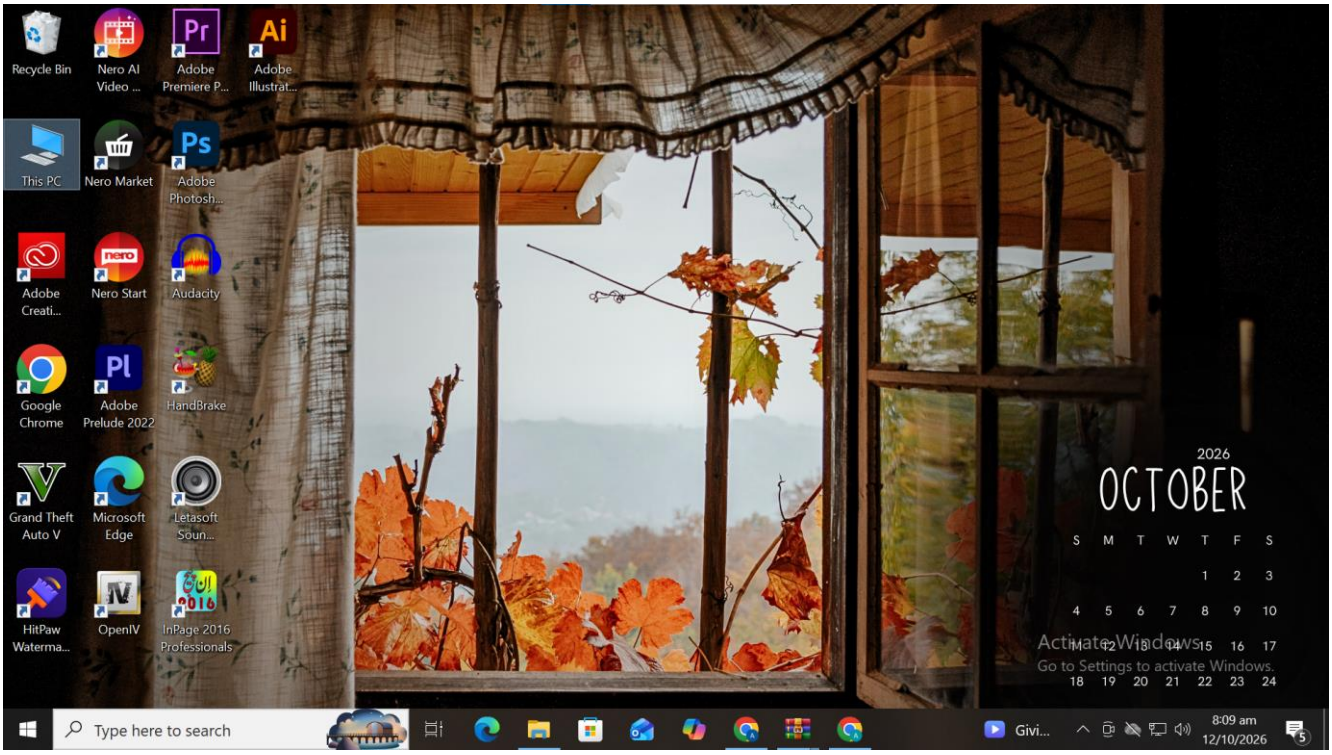
I explored the Windows 93 environment and observed its desktop, icons, taskbar, menus, and application windows. It provides a retro desktop experience through a web browser without requiring a traditional operating-system installation.

2. Compare the graphical user interfaces of Windows 98 and modern Windows operating systems

Objective: To compare the graphical user interface (GUI) of Windows 98 with modern Windows operating systems.



Windows 98



Window 10 pro

Comparison table

Feature	Windows 98	Modern Windows
Desktop design	Simple, classic appearance	More modern design with updated visual effects
Start menu	Traditional Start menu	Updated Start menu and search features
Taskbar	Basic taskbar for open applications	Improved taskbar with additional features
File management	Classic Windows Explorer	Updated File Explorer and improved navigation
Settings	Control Panel with basic options	Settings app and Control Panel, depending on version
Security	Limited security protections by modern standards	Improved security features, depending on version and configuration
Hardware support	Designed for older hardware	Supports newer hardware and technologies
Internet features	Older browser and networking technologies	Modern networking and browser support

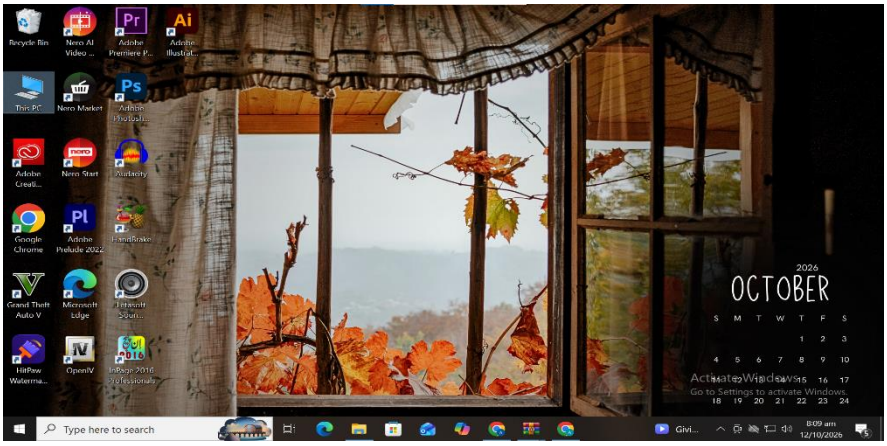
Windows 98 has a simple, classic interface with a Start button, taskbar, desktop icons, and traditional menus. Modern Windows versions provide an updated appearance, improved search, enhanced security features, and better support for current hardware and software.

3. Explain the difference between a graphical user interface (GUI) and a command-line interface (CLI)

A Graphical User Interface (GUI) allows users to interact with a computer through visual elements such as windows, icons, buttons, and menus. A Command-Line Interface (CLI) allows users to interact with a computer by typing commands into a terminal.

GUI — Graphical User Interface

- Uses icons, menus, buttons, and windows.
- Easy for beginners to learn.
- Users usually interact using a mouse and keyboard.
- Example: Opening a folder using File Explorer.



CLI — Command-Line Interface

- Uses typed commands.
- Requires knowledge of command syntax.
- Can perform tasks quickly and automate repetitive work.
- Example: Typing `hostname` in Command Prompt to display the computer name.

```
C:\Users\umair>hostname
DESKTOP-NHKHR1N
```

Difference between GUI and CLI

Feature	**GUI**	**CLI**
Interaction	Clicking visual elements	Typing commands
Ease of use	Generally easier for beginners	Requires command knowledge
Speed	Convenient for visual tasks	Often faster for repetitive tasks
Resource usage	May require more graphical resources	Often uses fewer graphical resources
Example	File Explorer	Command Prompt or Linux Terminal

During this practical, I explored a browser-based Windows 93 environment, learned about the classic Windows 98 interface, and compared older desktop designs with modern Windows. I also learned that a GUI uses visual elements, while a CLI uses typed commands to interact with the operating system.

THE END